

Geometric Assessment of a Large Component Jump in the Reconstructed Plastic Tensor

Abstract

A large component change in the reconstructed plastic tensor $T = F_P H$ was observed in element 389 during a plastic-well transition. In particular, T_{21} changed from 17 to 27, while several other entries also changed by order-one or larger integer amounts. This event is not a numerical bug. The component values are large because the corresponding wells lie close to the boundary of the Poincare disk, where Euclidean matrix components are a strongly distorted measure of geometric separation. When the pre-transition state is moved to the centre by an exact hyperbolic isometry, the post-transition state is at the nearest-neighbour well distance

$$d_H = 0.962423650119 = d_{\text{nn}}.$$

The complete saved trajectory contains eleven distinct states and ten recorded changes. Nine recorded changes are single nearest-neighbour moves. The only coarsely sampled exception factors exactly into two nearest-neighbour shear moves with one intermediate well absent from the saved output. Thus the trajectory is consistent with local motion on the well graph, and the resolved large jump in element 389 is precisely a transition to an adjacent well.

1 Observation and diagnostic question

The reconstructed tensor

$$T = F_P H$$

was introduced so that plastic history can be transported through mesh reconnection without inheriting an unphysical dependence on the current finite element connectivity. During a detailed minimization sequence for element 389, the following transition was isolated:

$$T_0 = \begin{pmatrix} 27 & 19 \\ 17 & 12 \end{pmatrix} \longrightarrow T_1 = \begin{pmatrix} 10 & 7 \\ 27 & 19 \end{pmatrix}. \quad (1)$$

The change $T_{21} : 17 \rightarrow 27$ is large when judged component by component. Component continuity is not, however, the relevant physical criterion for a periodic tensorial energy landscape. Distinct integer matrices label different wells, and their entries can become arbitrarily large even when the wells remain adjacent in the underlying hyperbolic geometry.

The appropriate diagnostic question is therefore not whether the entries of T remain small. It is whether the transition in

$$C_T = T^\top T$$

connects neighbouring wells in the Poincare representation. A transition that skips non-neighbouring wells during a fully resolved minimization event would indicate a possible defect in the update rule. A transition to an adjacent well is the expected local plastic event, irrespective of the size of the matrix entries used to label the two wells.

2 Poincare-disk representation

For a symmetric positive-definite matrix C , define

$$\hat{C} = \frac{C}{\sqrt{\det C}}.$$

Writing the components of $\hat{C}$ as $\hat{C}_{11}, \hat{C}_{12}, \hat{C}_{22}$, the disk coordinates used here are

$$x_p = \frac{\hat{C}_{11} - \hat{C}_{22}}{2 + \hat{C}_{11} + \hat{C}_{22}}, \quad y_p = \frac{2\hat{C}_{12}}{2 + \hat{C}_{11} + \hat{C}_{22}}. \quad (2)$$

Let $z = x_p + iy_p$. The hyperbolic distance between two disk points is

$$d_H(z_0, z_1) = 2 \operatorname{artanh} \left| \frac{z_1 - z_0}{1 - \bar{z}_0 z_1} \right|. \quad (3)$$

This distance, rather than the Euclidean norm of $T_1 - T_0$, measures the separation of the two wells.

The trajectory in Fig. 1 was obtained by extracting the four components of T for element 389 from all 347 saved simulation states and 62 detailed minimization states around the event in Eq. (1). Repeated consecutive matrices were removed for plotting, leaving eleven distinct retained states. The connecting lines show the order of the retained states; they are not intended as Euclidean interpolations of the physical trajectory.

3 The resolved large jump is a nearest-neighbour transition

The geometry of the highlighted event is most transparent after moving the pre-transition state to the centre of the disk. This can be done directly in matrix form. Define

$$M = T_0^{-1} = \begin{pmatrix} 12 & -19 \\ -17 & 27 \end{pmatrix}, \quad \tilde{T} = TM, \quad \tilde{C} = M^\top C_T M. \quad (4)$$

The pre-transition state then satisfies $\tilde{T}_0 = T_0 T_0^{-1} = I$, so it maps exactly to the disk origin. Because the congruence in Eq. (4) is an isometry of the determinant-one positive-definite manifold, it preserves hyperbolic distances and well adjacency.

For the post-transition state,

$$\tilde{T}_1 = T_1 M = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad \tilde{C}_1 = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5)$$

Equation (2) therefore gives

$$\tilde{z}_1 = 0.2 - 0.4i, \quad |\tilde{z}_1| = \frac{1}{\sqrt{5}}. \quad (6)$$

The distance from the centred pre-transition state is consequently

$$d_H(0, \tilde{z}_1) = 2 \operatorname{artanh} \left(\frac{1}{\sqrt{5}} \right) = 0.962423650119. \quad (7)$$

A unit integer shear, for example $R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, has exactly the same distance from the identity well. Hence

$$d_H(T_0, T_1) = d_{\text{nn}},$$

and the large component jump is exactly one nearest-neighbour transition. Figure 2 displays the same conclusion geometrically: the pre-transition state is at the origin and the red arrow terminates at the adjacent well selected by the Poincare grid.

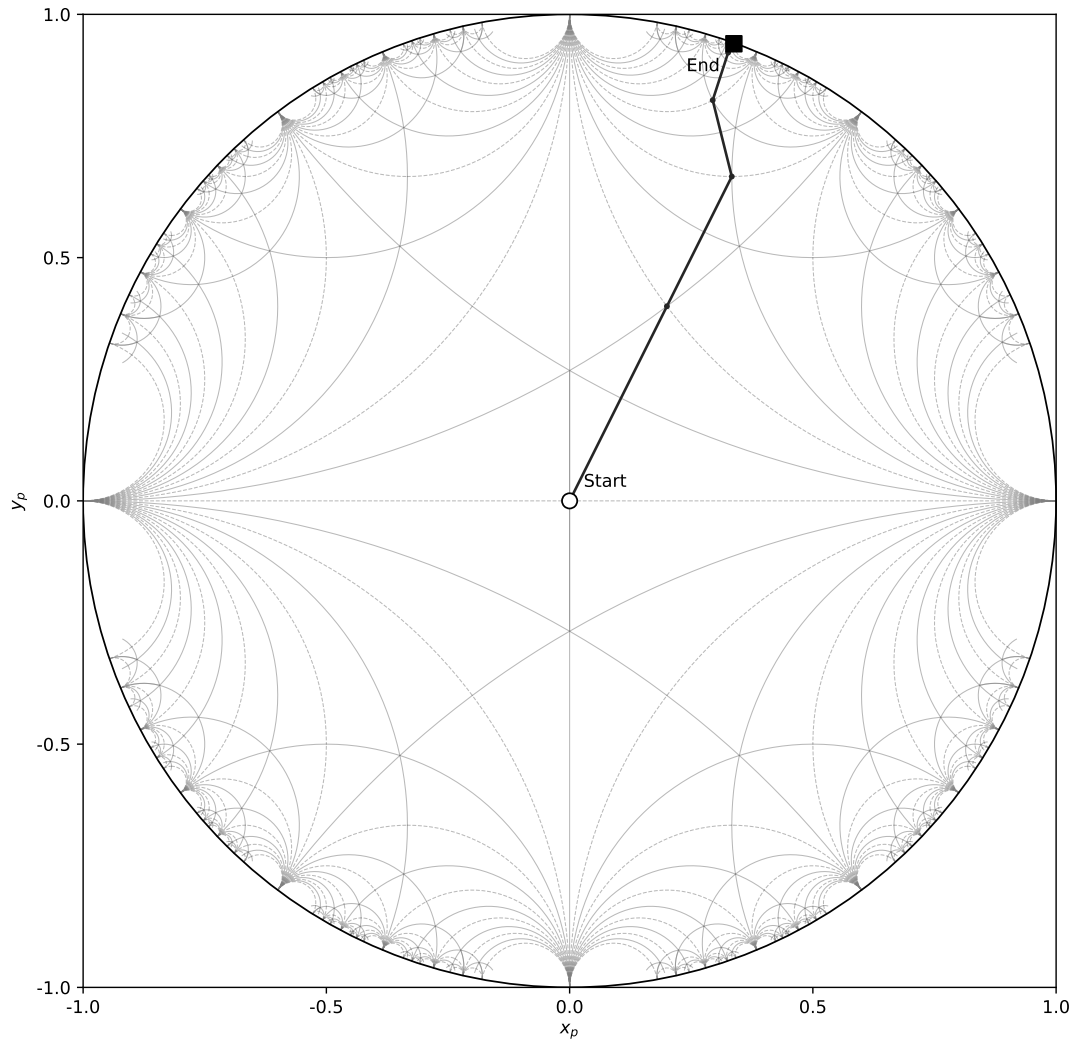


Figure 1: Complete retained Poincare-disk path of T for element 389. The matrix components grow along the trajectory as the state approaches the disk boundary, but the path continues to follow the local well structure.

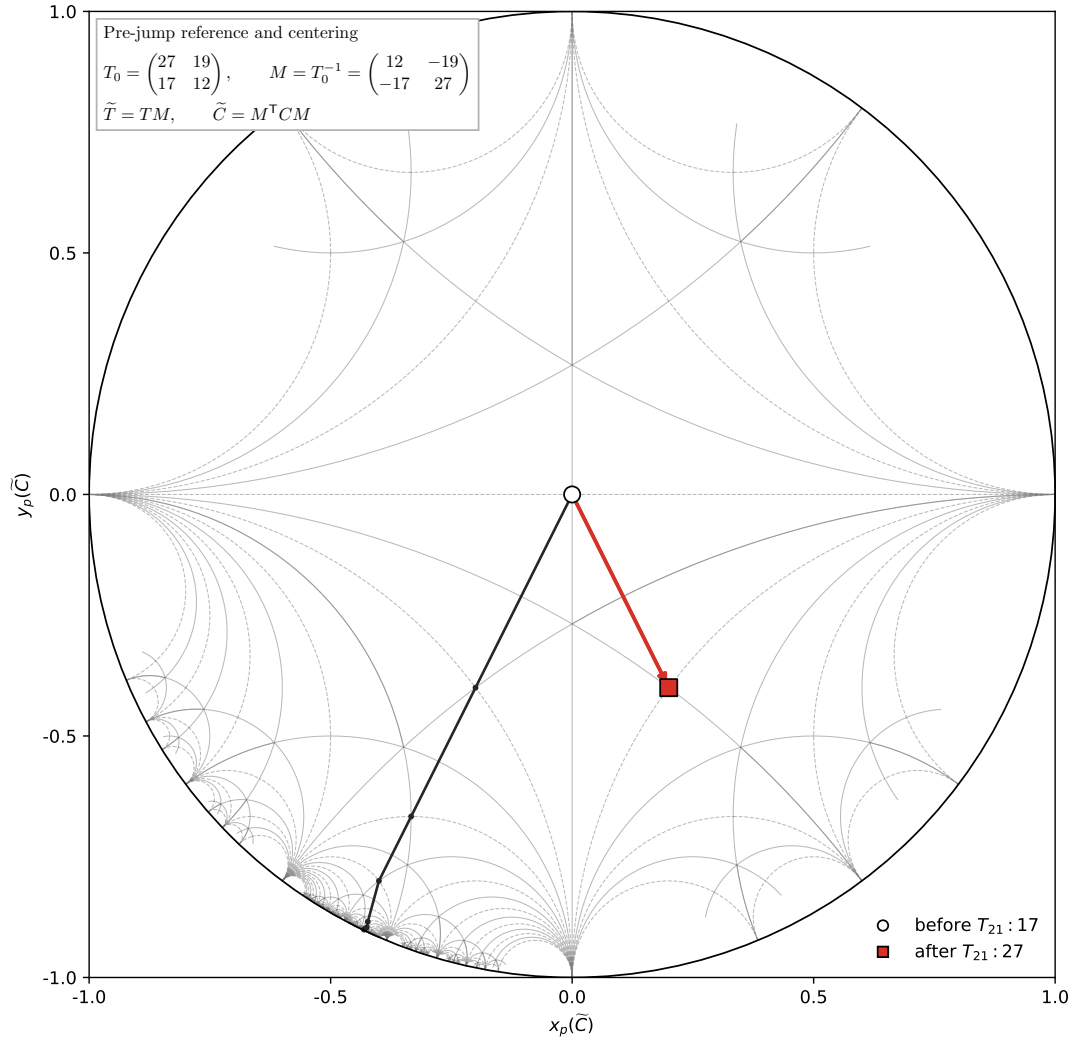


Figure 2: The full trajectory after the matrix centering transformation $M = T_0^{-1}$. The open circle is the pre-transition state and the red square is the post-transition state. The transformation preserves the Poincaré grid and makes the nearest-neighbour character of the highlighted event directly visible.

4 Neighbour structure of the complete retained path

The hyperbolic distances were also evaluated for every consecutive pair among the eleven distinct retained matrices. The result is summarized in Table 1.

Quantity	Count
Distinct retained states	11
Consecutive retained changes	10
Changes at exactly one nearest-neighbour distance	9
Coarsely sampled intervals factoring into two neighbour moves	1
Resolved nonlocal jumps	0

Table 1: Classification of the retained element-389 trajectory.

The single retained chord that is not one nearest-neighbour distance occurs between

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix}.$$

It is not an incompatible nonlocal state change. With the unit-shear generator

$$U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

the endpoints satisfy

$$A \longrightarrow UA = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix} \longrightarrow U^2A = B. \quad (8)$$

Thus the saved chord has well-graph distance two and admits an exact nearest-neighbour intermediate state. The detailed minimization states for this earlier load interval were not saved, so the two constituent moves cannot be temporally resolved from the available files. The appropriate conclusion is that every resolved event is local, while this one coarse interval is consistent with two local events between outputs. In particular, the event that motivated the present investigation is resolved in detail and has the single-neighbour distance in Eq. (7).

5 Interpretation and consequence for the update rule

The observed component jump does not provide evidence that T is being updated incorrectly. The Poincare-disk boundary corresponds to increasingly anisotropic determinant-one metrics. Integer representatives of nearby wells can therefore contain large entries, and an adjacent-well transition can replace several large components simultaneously. Component differences such as $|T_{21}^+ - T_{21}^-|$ are consequently unsuitable as a locality or continuity criterion.

For this event, changing the T - or H -update rule in order to suppress the component jump would remove a geometrically valid plastic transition. A more reliable diagnostic for future cases is:

1. construct $C_T = T^T T$ for the element before and after the event;
2. centre the pre-event state with $M = T_0^{-1}$, or compute the invariant distance directly from Eq. (3);
3. compare the result with the nearest-neighbour distance d_{nn} and inspect the well-graph factorization;

4. use minimization-resolved output whenever a saved interval spans more than one graph edge.

This criterion is invariant under the large integer labels that arise near the disk boundary and directly tests the physically relevant structure of the periodic energy landscape.

6 Conclusion

The large change in the components of T for element 389 is not a bug. The resolved transition

$$\begin{pmatrix} 27 & 19 \\ 17 & 12 \end{pmatrix} \longrightarrow \begin{pmatrix} 10 & 7 \\ 27 & 19 \end{pmatrix}$$

has exactly the hyperbolic distance of one unit-shear neighbouring well. The complete retained path likewise consists of nearest-neighbour moves, apart from one coarse output interval that factors exactly into two such moves. Large integer component changes are therefore a coordinate effect of approaching the Poincare-disk boundary, not evidence of a nonphysical jump in the plastic state.